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A Comparison of Treatment Outcomes after Standard Dose (70 Gy) versus Reduced Dose (50 Gy) Proton Radiation in Patients with Uveal Melanoma

Anne Marie Lane, MPH,[1] Monica M. Oxenreiter, MPP, MBA,[1] Mustafa Hashmi, BS,[1] Mary E. Aronow, MD,[1] Alexei V. Trofimov, PhD,[2] Helen A. Shih, MD,[2] Evangelos S. Gragoudas, MD,[1] Ivana K. Kim, MD[1]

Objective: To compare outcomes in a large patient cohort with small-medium tumors located within 1 disc diameter (DD) of the optic nerve and/or fovea treated with 50 Gy or 70 Gy proton therapy.
Design: Retrospective cohort study.
Subjects: A total of 1120 patients with uveal melanomas $\leq$ 15 mm in largest basal diameter, $\leq$ 5 mm in height, located within 1 DD of the optic nerve and/or fovea, who received primary treatment with protons between 1975 and 2016 at Massachusetts Eye and Ear/Massachusetts General Hospital.
Methods: The rates of outcomes were estimated using the Kaplan-Meier method. Differences between the radiation dose groups were tested using the log-rank test.
Main Outcome Measures: Local tumor recurrence, melanoma-related mortality, and visual acuity preservation ($\geq$ 20/200, $\geq$ 20/40).
Results: Local tumor recurrence was observed in 1.8% of the 50 Gy group and 1.5% of the 70 Gy group. The median time to recurrence was 30.7 months for patients treated with 50 Gy and 32.0 months for those treated with 70 Gy ($P = 0.28$). Five-year rates of vision retention ($\geq$20/40, $\geq$ 20/200) were 19.4% and 49.3% for patients treated with 50 Gy and 16.4% and 40.7% in those treated with 70 Gy. Ten-year rates of melanoma-related mortality were 8.4% in the 50 Gy group and 8.9% in the 70 Gy group ($P = 0.47$).
Conclusions: Comparable rates of local control are achieved treating small-medium tumors near the optic nerve and/or fovea with 50 Gy or 70 Gy proton therapy, supporting the use of the lower dose in patients with these tumor characteristics. *Ophthalmology Retina 2022;6:1089-1097 © 2022 by the American Academy of Ophthalmology*

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Proton radiation is an effective treatment for patients with uveal melanoma. Rates of local tumor control and eye retention are high, and retention of useful vision is achieved in many patients.[1−4]

Importantly, the properties of protons allow for the delivery of a maximum density of ionization in a sharply focused, localized volume (Bragg peak). The uniform dose of radiation delivered to the entire tumor and the sharp reduction of the dose outside the treated area make it an attractive treatment modality for tumors located near the optic nerve or fovea because radiation-induced damage to normal ocular tissue can be reduced.

Since the first patient with uveal melanoma was treated with proton therapy at the Harvard Cyclotron Laboratory in 1975, the standard total dose of proton irradiation used to treat patients with uveal melanoma at our center has been 70 Gy given in 5 fractions, chosen based upon clinical studies of patients with cutaneous melanoma.[5,6] Fractionation schemes employed at other proton facilities include delivering between 56 Gy and 60 Gy in 4 fractions.[7] Although high rates of local control are achieved, significant ocular morbidity occurs, particularly for patients whose tumors are located near the fovea or optic nerve.[8] Patients with such tumors are at a high risk for vision loss from radiation maculopathy[9] and papillopathy.[10−12] Dose-incidence curves reveal that risk of papillopathy approaches 100% when the optic disc receives the full dose of 70 Gy, whereas the risk of maculopathy increases with any radiation exposure and levels after ∼ 40 Gy.[12]

Published studies of visual outcomes for peripapillary tumors treated with proton radiation have reported rates of visual acuity (VA) retention of $\geq$ 20/200 ranging from 14% to 43% at 5 years after treatment.[10,13,14] For tumors within 1 disc diameter (DD) of the fovea, 36% of the patients were found to have a VA of $\geq$ 20/200 at 5 years after proton radiation.[9]

To address the deleterious effects of radiation in such tumors, we previously completed a radiation dose reduction

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